

IBERET ACTIVE FILMTAB

Controlled-Release Iron with Vitamin C and Vitamin B-Complex

DESCRIPTION

Iberet Active Filmtab contains iron in a controlled-release vehicle which is an inert, porous, plastic matrix. The ferrous sulphate is impregnated in this matrix, called the Gradumet. Ascorbic acid and the B-complex vitamins are layered on the Gradumet. The vitamins and mineral are leached from the Gradumet as it passes through the gastrointestinal tract. Then the empty matrix is excreted harmlessly in the stool.

Iberet Active Filmtab is supplied as red, oval-shaped tablets.

Each tablet of Iberet Active Filmtab provides the following:

Ingredient	
Ferrous Sulfate (equivalent to 105 mg of elemental iron)	525 mg
Vitamin C (as sodium ascorbate)	500 mg
Niacinamide	30 mg
Calcium pantothenate	10 mg
Vitamin B ₁ (thiamine mononitrate)	6 mg
Vitamin B ₂ (riboflavin)	6 mg
Vitamin B ₆ (pyridoxine hydrochloride)	5 mg
Vitamin B ¹² (cyanocobalamin)	25 mcg

INDICATIONS AND USAGE

For prevention and / or treatment of vitamin deficiencies.

DOSAGE AND ADMINISTRATION

Iberet Active Filmtab is administered orally and may be taken on an empty stomach.

Usual adult dose, including pregnant females: One tablet daily, or as directed by a physician.

Effectiveness of iron therapy should be monitored via standard haematology tests.

Do not chew or crush tablet, swallow whole.

Route of administration: oral

CONTRAINDICATIONS

Iberet Active Filmtab is contraindicated in individuals known to be hypersensitive to any of its components.

Iron therapy is contraindicated in patients with thalassemia, sideroblastic anemia, hemochromatosis and hemosiderosis.

Repeated blood transfusions or parenteral iron therapy contraindicate concurrent oral iron therapy.

Intestinal diverticula or any intestinal obstruction contraindicate use of the Gradumet matrix.

Niacinamide

This is contraindicated in patients with liver disease or active peptic ulcer. Niacin is also contraindicated in patients with arterial hemorrhaging or severe hypotension.

This product should not be used in pediatric patients.

WARNINGS AND PRECAUTIONS

General

Patients should be advised to keep these products out of the reach of children.

Acute iron poisoning occurs rarely in adults, but it could happen if children swallow this medication. See the **OVERDOSAGE** section for symptoms and treatment of acute iron poisoning.

Accidental overdose of iron-containing products is a leading cause of fatal poisoning in children under six years of age. Keep this product out of the reach of children. In cases of accidental overdose, call a doctor or seek medical advice immediately.

Iron therapy could induce relapse of erythropoietic protoporphyria. Iron overload has been suggested as being involved in the pathogenesis of porphyria cutanea tarda.

Where anemia exists, its nature should be established and underlying cause determined. As is the case with most medications, these should be stored out of reach of children.

Concomitant parenteral therapy with vitamin B₁₂ may be necessary in patients with deficiency of vitamin B₁₂.

Caution should be exercised when dosing ascorbic acid in patients with chronic renal failure and in patients receiving acetylsalicylic acid.

DRUG INTERACTIONS

Alcohol

Niacin with concomitant use of alcohol, toxic delirium and lactic acidosis has been noted.

Nicotine

Niacin with concomitant use of nicotine; increased flushing and dizziness have been reported.

Tetracyclines

Iron inhibits absorption of tetracyclines from the GI tract and vice versa. If both drugs must be given, administer tetracycline three hours after or two hours before oral iron supplements.

Chloramphenicol

Chloramphenicol may delay response to iron therapy, and may diminish response to B₁₂-deficient patients; alternate anti-infective therapy should be considered.

Antacids

Antacids may decrease iron absorption.

Cholestyramine

Cholestyramine may decrease iron absorption.

Quinolone Anti-infective Agents

Concurrent administration of oral iron preparations may interfere with the oral absorption of some quinolone anti-infective agents (e.g. ciprofloxacin, norfloxacin, ofloxacin) resulting in decreased serum and urine concentrations of the quinolones. Therefore, oral iron preparations should not be ingested with or within two hours of a dose of an oral quinolone.

Penicillamine

Iron decreases the cupruritic effect of penicillamines. If both drugs must be administered, then administration should be at least two hours apart.

Anti-diabetic agents

Dosage requirements for these agents may change while taking niacinamide.

Ganglionic Blocking Drugs

Niacin may potentiate the hypotensive effect of these agents.

Neuromuscular Blocking Agents

Thiamine may enhance the effect of these agents.

Propantheline Bromide

Prior administration of propantheline bromide may delay the absorption time but increase the absorption concentration of riboflavin.

Levodopa

Since pyridoxine is noted to have effects on dopamine, drug interactions are possible. A drug interaction with levodopa is noted, but can be avoided if levodopa is given in combination with a decarboxylase inhibitor.

Phenobarbital and Phenytoin

High doses of pyridoxine hydrochloride (greater than the 5 mg/day from Iberet-500) have been reported to decrease serum concentrations of phenobarbital or phenytoin.

Vitamin B₁₂ Absorption From the GI Tract Can Be Diminished By Various Agents

Aminoglycoside antibiotics, colchicine, extended-release potassium preparations, aminosalicic acid and its salts, anticonvulsants, cobalt irradiation of the small bowel, or chronic excessive alcohol intake. In older patients, serum B₁₂ levels should be regularly assessed during treatment with Iberet Active.

Anticoagulants

Prothrombin times are decreased when ascorbic acid is used concomitantly with anticoagulants.

Drug Food Interactions

Absorption of iron is inhibited by ingestion of eggs, milk, phytate-containing substances such as bran, or tannins in tea.

Oral iron preparations should not be taken within one hour or two hours after ingestion of the above mentioned products.

Laboratory Tests

Iron

Iron preparations may turn stools black, which could hinder detection of occult blood in feces. The guaiac test also occasionally yields false-positive tests.

Ascorbic acid

As a strong reducing agent, ascorbic acid may interfere with tests based on oxidation-reduction reactions. Ascorbic acid in the urine may produce false results in glucosuria determinations. Prothrombin times are decreased when ascorbic acid is used concomitantly with anticoagulants.

Niacin

Niacin may cause false elevations in fluorometric tests of urinary catecholamines. It may also give false positive results when a cupric sulfate solution (such as Benedict's reagent) is used for urinary glucose determination.

Thiamine and Pyridoxine Hydrochloride

Thiamine may cause false positive results in the phosphotungstate method for determination of uric acid. Thiamine or pyridoxine hydrochloride may cause false positive results for urobilinogen in the spot test using Ehrlich's reagent.

Riboflavin

Large doses of riboflavin may result in bright yellow urine, which may interfere with urinalysis based on spectrometry or color reactions. Riboflavin may also cause false elevations in fluorometric determinations of catecholamines or urobilinogen.

Vitamin B₁₂

Methotrexate, pyrimethamine, and most anti-infectives invalidate diagnostic microbiologic blood assays for vitamin B₁₂. Prior administration of this vitamin may result in false-positive test results for IF antibodies, which are present in approximately 50% of patients with pernicious anemia.

PREGNANCY AND LACTATION

Nursing Women

B-complex vitamins diffuse across the placenta and also appear in breast milk.

ADVERSE REACTIONS

Reported side effects are similar to those associated with conventional oral iron preparations i.e., nausea, vomiting, abdominal pain or discomfort, diarrhea or constipation. However, the incidence of side effects is lower due to the controlled-release nature of the formulation. If such reactions occur, the tablet may be taken after a meal.

Allergic reactions, including rash, pruritus, and anaphylaxis have been reported with vitamin use.

Components of products have been associated with gastrointestinal effects such as heartburn, eructation, abdominal pain and cramps, diarrhea, vomiting nausea, and anorexia.

Hepatic dysfunction with abnormal liver function tests, including hyperbilirubinemia has been noted.

Deterioration of acneform vulgaris, or eruption of acneform exanthema, has been noted with several components.

Bright yellow urine discoloration has been reported with riboflavin usage.

Niacinamide has strong vasodilator effects, most often characterized by flushing, dizziness or faintness.

Peripheral sensory neuropathies have been noted with usage of pyridoxine.

Stone formation, crystalluria, and oxalosis, with ascorbic acid usage, have been reported in the literature.

Black discoloration of the stool has been reported with iron usage.

Isolated cases of injury to mouth and pharynx, oesophageal ulcer, haemataemesis, and ileus have been reported.

Acute iron intoxication may result in increased capillary permeability, reduced plasma volume, increased cardiac output, and sudden cardiovascular collapse.

Ascorbic acid is usually well tolerated. Large doses are reported to cause diarrhea and other gastrointestinal disturbances and are associated with the formation of renal calcium oxalate calculi.

OVERDOSAGE

Due to the controlled-release nature of the preparation, effects of overdosage could be delayed.

Acute overdosage of iron may cause nausea and vomiting and, in severe cases, cardiovascular collapse and death. The lethal dose of orally ingested elemental iron is estimated to be 180 to 300 mg/kg of body weight. However, a dose of elemental iron as low as 30 mg/kg may be toxic in some individuals and ingestion of doses as low as 60 mg/kg have resulted in death. In small children, however, it is estimated that as little as 400 mg of elemental iron could be fatal.

Toxicity that occurs with an acute iron overdosage results from a combination of the corrosive effects on the gastrointestinal mucosa and the metabolic and hemodynamic effects caused by the presence of excessive elemental iron.

The signs and symptoms of acute iron poisoning may occur within 10 to 60 minutes or be delayed for several hours. Initial clinical manifestations may encompass acute gastrointestinal irritation, including epigastric pain, nausea, vomiting, diarrhea of green and subsequently tarry stools, melena and hematemesis which may be associated with drowsiness, pallor, cyanosis, lassitude, seizures, shock, and coma. Because of potential toxic effects of overdosage, immediate medical attention is warranted.

Iron poisoning should be treated by emptying the stomach via ipecac-induced vomiting or preferably, by gastric lavage with a large bore tube. If the patient has experienced

multiple episodes of vomiting, especially if the vomitus contains blood, ipecac syrup should not be administered.

Vomit should be examined for returned Gradumet tablets. If sufficient tablets are not returned, the possibility of whole gut lavage with 0.9% sodium chloride solution plus a saline cathartic should be considered. Surgical removal of iron tablets which are visible in abdominal radiographs may be required if other means of removing the drug are unsuccessful.

The best method for assessing the severity of an iron ingestion is to measure the serum iron and the total iron binding capacity (TIBC). If the serum iron level is greater than TIBC, the potential for systemic toxicity exists. Serum iron and total iron-binding capacity levels may be used as guidelines for use of deferoxamine, an agent used to chelate elemental iron.

Chelation therapy with deferoxamine should be considered when the following conditions exist:

1. a potentially lethal dose (180 to 300 mg/kg or more) of elemental iron has been ingested;
2. serum iron concentrations are greater than 400 to 500 µg/dL.
3. serum iron concentration exceed total iron binding capacity, and/or;
4. patients have severe symptoms of iron intoxication such as coma, shock, or seizure.

Hemodialysis is of little value in the treatment of iron intoxication.

Supportive treatment, including suction and maintenance of airway; correction of acidosis and control of shock and dehydration with intravenous fluids or blood, oxygen and vasopressors, should be administered as required.

High doses of individual components of the product have been associated with eczematous and exanthematous skin lesions, fatigue, and insomnia.

In higher niacinamide doses, liver damage, gout and ulcer formation have been noted. Vasodilatory effects such as giddiness, faintness, vasovagal attacks, and anaphylactic shock have also been reported.

In high doses of pyridoxine, peripheral sensory neuropathy and vesicular skin lesions have been reported.

Hemolysis has been reported at high doses of ascorbic acid, especially in patients with glucose 6 phosphate dehydrogenase deficiency.

CLINICAL PHARMACOLOGY

Pharmacodynamics

Iron, an essential mineral, is a component of hemoglobin, myoglobin and a number of enzymes. The total body content of iron is approximately 50 mg/kg in man and 35 mg/kg in women.

Iron is primarily stored in the body as hemosiderin or ferritin, found in the reticuloendothelial cells of the liver, spleen, and bone marrow. Approximately two-thirds of total body iron is in the circulatory red blood cell mass in hemoglobin, the major factor in oxygen transport. Concentration of plasma iron and the total iron-binding capacity of plasma vary greatly in different physiological conditions and disease states.

Pharmacokinetics

Absorption

The absorption is increased when iron stores are depleted or red blood cell production is increased. Conversely, high iron blood concentrations decrease absorption. The average dietary intake of iron is 18 to 20 mg/day. Approximately 10% of this iron is absorbed in healthy individuals and about 20% to 30% in iron-deficient individuals.

Folic acid and iron are absorbed in the proximal small intestine, particularly the duodenum. Folic acid is absorbed maximally and rapidly at this site, and iron is absorbed in a descending gradient from the duodenum distally. After absorption, folic acid is rapidly converted into its metabolically active forms. Except for the folates ingested in liver, yeast, and egg yolk, the percentage of absorption of food folates averages about 10%.

The ferrous salt form is absorbed three times more readily than the ferric form. The common ferrous salts (sulfate, gluconate, fumarate) are absorbed almost on a milligram-for-milligram basis, but differ in the content of elemental iron. Ferrous sulfate comprises 20% of elemental iron content.

Oral iron is absorbed most efficiently when it is administered between meals. However, conventional iron preparations frequently cause gastric irritation when taken on an empty stomach. Although food can decrease the absorption of iron by 40% to 66%; gastric intolerance may necessitate administering the drug with food.

Studies with iron in the Gradumet have indicated that relatively little iron is released in the stomach, gastric intolerance is seldom encountered, and hematologic response ranks with that obtained from plain ferrous sulfate. Therefore, potential gastric irritation is minimized when iron is administered in the Gradumet form in comparison with conventional oral iron preparations.

Large amounts of ascorbic acid administered orally with ferrous sulfate have been shown to enhance iron absorption. This is apparently due to the ability of ascorbic acid to prevent the oxidation of ferrous iron to the less effectively absorbed ferric form.

The B-Complex vitamins are absorbed by an active transport process; they are rapidly eliminated and therefore are not stored in the body.

Calcium pantothenate is absorbed readily from the gastrointestinal tract and distributed to all body tissues.

Distribution

Ferrous iron passes through gastrointestinal mucosal cells directly into the blood and is

immediately bound to transferrin. Transferrin, a glycoprotein β_1 -globulin, transports iron to the bone marrow where it is incorporated into the hemoglobin.

Small excesses of iron within the villous epithelial cells are oxidized to the ferric state. Ferric iron combines with the protein apoferritin to yield ferritin and is stored in mucosal cells which are exfoliated at the end of their life span and excreted in the feces.

Elimination

Iron metabolism occurs in a virtually closed system. The majority of iron liberated by destruction of hemoglobin is conserved and reused by the body. The daily excretion of iron from urine, sweat and sloughing of intestinal mucosal cells amounts to approximately 0.5 to 1 mg in healthy men and 1 to 2 mg in menstruating women. The half-life of ferrous sulfate is approximately six hours.

Storage Condition:

Store below 30°C.

Presentations:

Box of 10's (1 x 10's)

Box of 100's (10 x 10's)

Manufacturer:

PT. Abbott Indonesia

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Revision Date: July 2025